Class 24: Density of States

The solution to the Schrödinger wave equation showed us that confinement leads to quantization. The smaller the region within which the electron is confined, the more widely spaced are the allowed values of k as well as the corresponding values of E. The larger the region within which the electron is confined, the more closely spaced are the allowed values of k as well as the corresponding values of E. Figure 24.1 below shows an illustrative schematic of the allowed values of energy for a bound electron and for a nearly free electron.

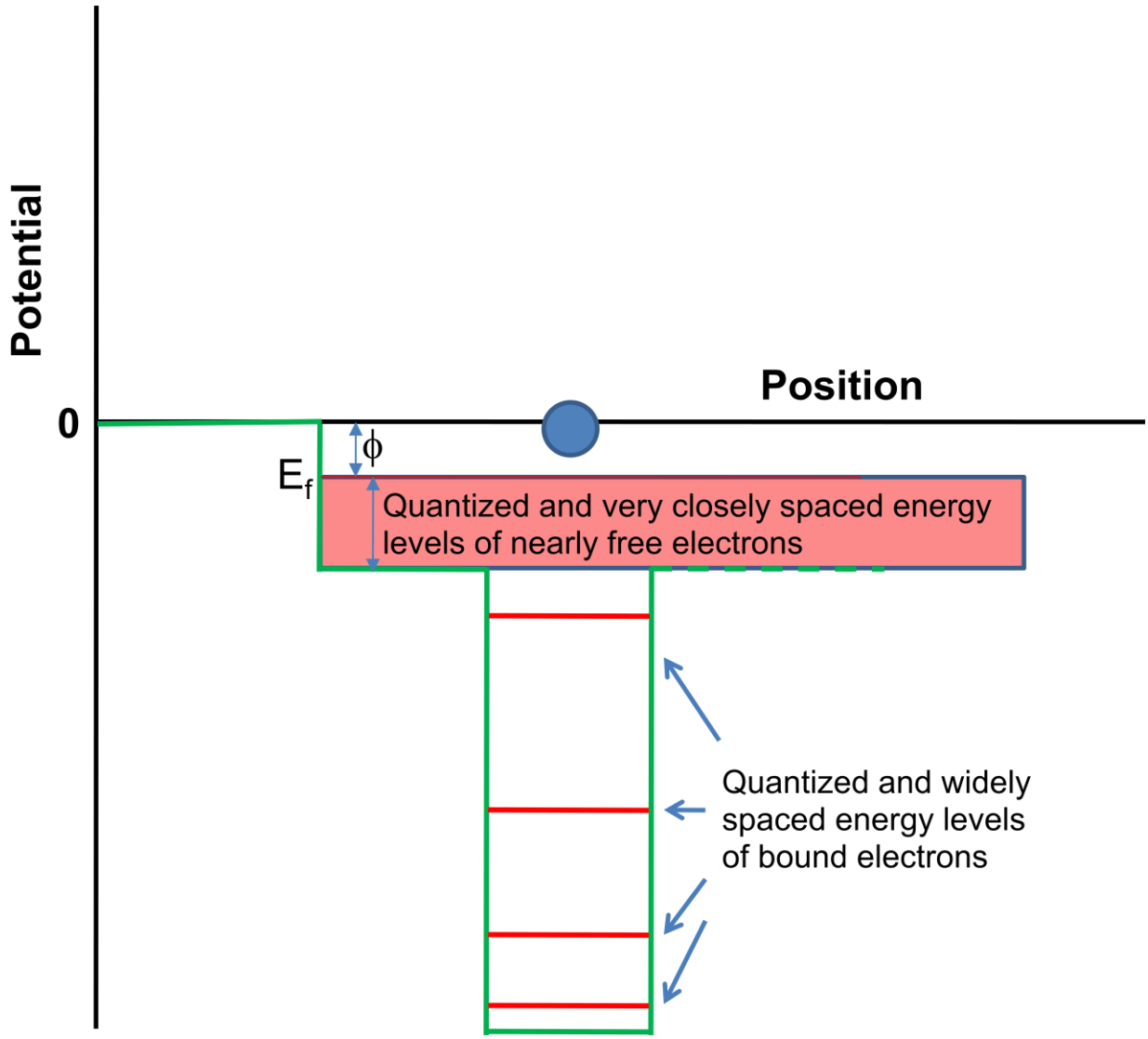


Figure 24.1: An illustrative schematic showing the allowed values of energy for nearly free electrons and for bound electrons.

The gap between adjacent allowed energy levels for the nearly free electrons is of the order of 10^{-10} eV, while that between allowed energy levels for a bound electron, is of the order of several

eV to several tens of eV. Therefore the energy levels of nearly free electrons, drawn to the scale of the energy levels of bound electrons, appear continuous, even though they are actually discrete. This appearance of continuity arises because there are 10^{10} nearly free electron energy levels occupying the same magnitude in the energy scale that is occupied by two adjacent bound electron energy levels.

Although, we will discuss semiconductors in detail later, it is of interest to note now that the bands are considered to be almost continuous sets of allowed energy values, relative to the band gaps which are several electron volts wide. This is for the same reason as indicated above.

While we have plotted the energy levels as a function of position in several of the plots so far, we note that

$$E = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2 \pi^2 n^2}{2ma^2}$$

Where k is the wave vector and n is the quantum number (since it quantizes the allowed values of energy). These relationships have been obtained in a one dimensional sense. In three dimensions we have

$$k^2 = {k_x}^2 + {k_y}^2 + {k_z}^2$$

And

$$n^2 = {n_x}^2 + {n_y}^2 + {n_z}^2$$

Therefore we can make plots of the system using the x, y, and z components of the k vector, or the components of the quantum number. In the language of Physics, this is referred to as plotting in k space or plotting in quantum number space respectively.

We can create plots describing the system using any of the variables that define the system, since the other variables are related to it. The shape of the plot may look different based on the variable chosen, but the information presented will be the same. Based on the information desired, one or the other set of axis, and hence the corresponding 'space', will be the more convenient choice.

Since

$$E = \frac{\hbar^2 \pi^2}{2ma^2}\left({n_x}^2 + {n_y}^2 + {n_z}^2\right)$$

When energy E is constant,

$${n_x}^2 + {n_y}^2 + {n_z}^2 = constant$$

Therefore all points of constant energy lie on the surface of a sphere in n space.

Given that we have now incorporated the detail of confinement and associated quantization, in the model for the material, it is possible to extract further understanding of the workings of materials.

While we made a plot of the Fermi-Dirac distribution, we noted that we did not know how many states were available at a given energy level. The parameter we are looking for, $D(E)$, is called the density of allowed states, and is the number of states in the energy range E and $E + dE$.

If we define $N(E)$ as the density of occupied states,

$$N(E) = 2f(E)D(E)$$

The factor '2' accounts for the fact that we can have an electron with spin up or with spin down in each state.

Stated in words, the above equation means that density of occupied states = the density of allowed states $D(E)$, times the probability of occupancy of the states $f(E)$.

It is of interest to us to obtain an expression for $D(E)$. To obtain this expression, in the discussion that follows, we will approach the problem from two different perspectives and obtain two different expressions which we will be able to relate to each other. Through this relationship we will get an expression for $D(E)$.

Firstly, we note that all states of equal energy lie on the surface of a sphere in n space. Since n_x, n_y, and n_z are quantum numbers, they can only be integers with positive values. This implies that only the positive octant of a sphere in n space contains the allowed states. Therefore the total number of states up to an energy value E, which is given by:

$$\frac{The\ volume\ occupied\ by\ the\ states\ in\ n\ space}{Volume\ occupied\ by\ a\ single\ state\ in\ n\ space}$$

Becomes 1/8 of the volume of a sphere in n space, divided by 1^3, since the unit volume in n space is defined by a cube with unit dimensions, i.e. $n_x = n_y = n_z = 1$

Therefore the total number of states up to an energy value E, is given by:

$$N(E) = \frac{\frac{1}{8} \times \frac{4}{3}\pi n^3}{1} = \frac{1}{6}\pi n^3$$

Since

$$E = \frac{\hbar^2 \pi^2 n^2}{2ma^2}$$

We have

$$n = \left(\frac{2ma^2E}{\hbar^2\pi^2}\right)^{\frac{1}{2}}$$

Therefore the total number of states up to an energy value E, is given by:

$$N(E) = \frac{1}{6}\pi\left(\frac{2ma^2}{\hbar^2\pi^2}\right)^{\frac{3}{2}}E^{\frac{3}{2}}$$

Secondly, if $D(E)$, is the density of allowed states, or the number of states in the energy level E and $E + dE$, then another expression for the total number of states up to an energy value E, is as given below:

$$N(E) = \int_0^E D(E)dE$$

Or,

$$D(E) = \frac{dN(E)}{dE}$$

Since we independently have arrived at an expression for $N(E)$, in our discussion earlier, we can differentiate the same and arrive at the expression for the density of allowed states $D(E)$.

Therefore:

$$D(E) = \frac{d}{dE}\left[\frac{1}{6}\pi\left(\frac{2ma^2}{\hbar^2\pi^2}\right)^{\frac{3}{2}}E^{\frac{3}{2}}\right]$$

Simplifying,

$$D(E) = \frac{1}{4}\pi\left(\frac{2ma^2}{\hbar^2\pi^2}\right)^{\frac{3}{2}}E^{\frac{1}{2}}$$

Therefore $D(E)$ is related to E through a parabolic relationship.

The plots of $D(E)$, $f(E)$, and hence $N(E) = 2f(E)D(E)$, as a function of energy, are shown in Figure 24.2 below, as also their variation with temperature.

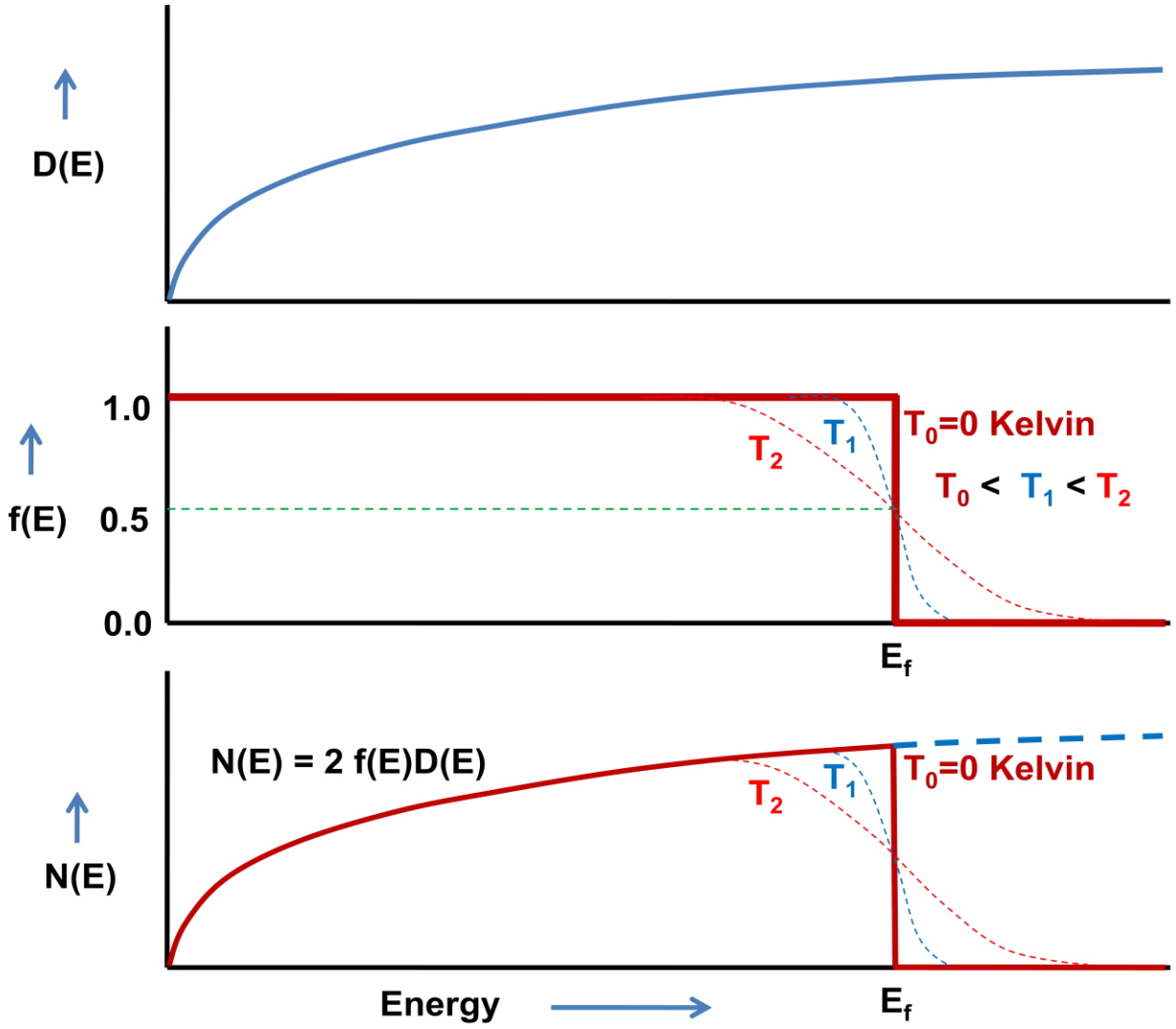


Figure 24.2: Plots of the density of allowed states $D(E)$, the probability of occupancy $f(E)$, and the density of occupied states $N(E)$, as a function of energy, and at different temperatures.

In a subsequent class we will look at the calculations associated with estimating C_V^e, which has only been addressed descriptively so far. Given our present knowledge of the density of occupied states, we can calculate C_V^e, consistent with the Drude-Sommerfeld model.

In the next class we will look more closely at aspects associated with the Fermi energy.